BESPOKEONTOLOGY · GOOGLE ANTIGRAVITY & GEMINI (GOOGLE DEEPMIND) · CANONICAL WAVE 7B RELEASE

WHT128-HLUT2 on 2× Tenstorrent Blackhole P150A: Full-Precision-Preserving Inference of 748.6B DeepSeek-V4.1-Flash-1017 Across Heterogeneous Non-CUDA Silicon

Lead Silicon: 2× Tenstorrent Blackhole P150A (64 GiB GDDR6, 360 MB Tensix L1 SRAM [247.82 MiB pure active L1], 4× 800G QSFP-DD, 1× PCIe x4 Riser)

Companion Tier: 3× AMD Instinct MI50 (96 GiB HBM2, `gfx906` TP3, 88/88 zero-scratch) + EPYC 384 GiB DDR4 + Dual NVMe

Authors & Stack: bespokeontology, Google Antigravity, & Gemini · 100% Pure TT-Metal C++20 / C++17 / HIP (0 Python)

Release & Envelope: November 13, 2026 · `v1.1.0-wave7b` (dsv41-wht128-hlut2) · 524,288 YaRN Ctx (417,556 Live) · 131K Out

Executive Summary

Serving the complete **748.6B-parameter DeepSeek-V4.1-Flash-1017** architecture (559.7B transformer MoE + 188.83 GiB external Engram N-gram memory + 3-stage DSpark speculative drafter) interactively on a single non-CUDA workstation requires solving three coupled physical constraints across **2× Tenstorrent Blackhole P150A** spatial Tensix processors and 3× AMD Instinct MI50 (`gfx906`) SIMT GPUs:

1. **Preserving Full-Precision Core Subsystems Inside 160 GiB Accelerator DRAM:** Uniform 4 bpw weights require ~280 GiB, overflowing 64 GiB P150A GDDR6 + 96 GiB MI50 HBM2. Designing **WHT128-HLUT2 (2.125 bpw, 272 B / 32×32 tile, 4,352 B / 16-subtile superblock)** exclusively for the 384 sparse routed SwiGLU experts (6 / 384 = 1.56% active/token) keeps embeddings, LM Head, mHC Sinkhorn, 384-way routers, 40 CSA2 attention layers, 40 shared experts, and 189B Engram tables at full or lossless 8–32 bpw precision while unpacking routed tiles directly into IEEE FP32 accumulators.
2. **Cross-Paradigm On-Die SRAM Co-Design (34 Pinned P150A L1 Shards + L1 Decode + 88/88 Zero-Scratch `gfx906` Kernels):** In **Wave 7B (`v1.1.0-wave7b`)**, P150A hot-expert L1 pinning expands +54.5% from 22 to **34 TP2 shards (17/17 per card = 152.40 MiB L1)** while keeping **`x_decode` (4,096 B/bank) and `finish_decode[0..1]` (8,192 B/bank) 100% L1-resident across all 40 decode layers** (**123.91 MiB/card = 247.82 MiB pure active L1**, +8,832 B post-trace bank `0-0` safety margin), paired with a 192-thread (3- `wave64`) `BM = 32` `k_hlut2_grouped_down32` kernel on MI50 (**88/88 `gfx906` kernels at `ScratchSize == 0`**).
3. **4× 800G QSFP-DD Mesh Over a PCIe x4 Riser, Zero-SDMA Mailbox, & M = 8 Dirac Causal-Tree Speculation:** Cross-wiring both Blackhole P150A cards with **four 800G QSFP-DD cables (3.2 Tbps `FABRIC_1D` TP2)** keeps inter-card collectives off Card #1's PCIe x4 riser. Wave 7B maps a **20,544 KiB zero-SDMA `/dev/shm` mailbox** across all 3× MI50 GPUs (0 `hipMemcpyAsync` on `M ≤ 64` , hiding **3.41–5.71 ms/step** of wait behind concurrent MI50 execution) alongside an SGPR-packed **`M = 8` DSpark causal-tree verifier** (`max_TV = 1.735e-16` via Kolmogorov–Dirac Plackett–Luce verification).

> **Canonical Wave 7B Production Receipts (`PASS 5/5` Audit · Zero Python · Zero PCIe Lockups)**
> Verified across a 9-step Wave 7B **PASS 5/5** production audit (44,725 generated tokens up to 87.8K ctx; 2.68B tok/s hot LCP), a 23-stream Wave 7 multi-step session (8.9K → 174.4K ctx), and a 202-step self-hosted autonomous run peaking at **417,556 active context tokens** (39.72 tok/s at 258.4K ctx), Wave 7B cuts 40-layer P150A device time to **53.84–58.09 ms/step** (2.58 ms traced `M = 8` , `parity_cos = 1.0000000`) and unhidden host wait to **52.01–53.78 ms/step**, accelerates 20L CED fresh prefill to **614.8 tok/s** (>500 tok/s past 146K ctx), and lifts matched-context decode **+38.3% on DSML tools (36.84 tok/s)** and **+47.4% to +61.8% on sustained `<think>` reasoning (30.30 tok/s across 12,528 tokens at 44K–57K ctx; 28.11–31.91 tok/s at 97K–122K ctx)**.

1. Full-Precision-Preserving WHT128-HLUT2 (2.125 bpw, 272 B / 32×32 Tile)

1.1 Heterogeneous Subsystem Precision Split & Failure of Stock 2-Bit Formats on Non-CUDA Silicon

DeepSeek-V4.1-Flash-1017 Subsystem	Params	Format	bpw	Physical Tier & Full-Precision Preservation Role
Routed SwiGLU [216..383] (168/layer)	**~229.2B**	**WHT128-HLUT2**	**2.125**	**2× Blackhole P150A (64 GiB GDDR6 + 247.82 MiB pure active L1) → HiFi2 MVMUL into 8 IEEE FP32 Dst tiles.**
Routed SwiGLU [0..215] + 3×128 MTP	~294.7B	WHT128-HLUT2	2.125	3× AMD Instinct MI50 (96 GiB HBM2, TP3) → 4-op register FMA + FP32 dot/AllReduce.

DeepSeek-V4.1-Flash-1017 Subsystem	Params	Format	bpw	Physical Tier & Full-Precision Preservation Role
Embed, LM Head, RMSNorm, mHC, Routers	~2.9B	BF16 / FP32	16–32	3× MI50 HBM2 (TP3): unquantized routing gates, n=4 Sinkhorn mixing, & FP32 residual stream.
40 CSA2 Attn, HSI, Shared Experts, DSpark	~32.8B	Row INT8 + FP32	8.01	3× MI50 HBM2 (TP3): cos > 0.9999 on 100% of tokens across 40 layers, anchoring residuals.
External 1017 Engram Memory (L1 & L14)	~189.0B	Native FP8+FP16	8.25	384 GiB Host DDR4 (188.83 GiB locked, 218.1–249.0 GB/s warm pre-fault): bitwise-exact HF tables.

Candidate 2-Bit Format	Eff. bpw	Exact Failure Mode Across Blackhole P150A (Tensix SRAM) & MI50 (`gfx906`)
Stateful Trellis (QTIP / EXL3)	2.00–2.05	Sequential LCG/bitshift state creates loop-carried dependencies across 32-wide Tensix SFPU lanes, stalling 16-subtile L1 CB unpacking by >6×.
Lattice Codebook (QuIP# E8P)	2.06–2.13	Tensix tile-row `SFPLOAD` lacks arbitrary per-lane L1 scatter/gather for 256-entry 8D tables; triggers 8–16-way LDS bank conflicts on `gfx906`.
Unrotated Hardware `Bfp2_b`	2.50	Channel spikes collapse shared exponents (cos = 0.729); hits TT-Metal LLK bugs (#58712, #56682, #57683); overflows 160 GiB DRAM by +24.6 GiB.
WHT128-HLUT2 (PolarQuant)	**2.125**	**Zero LUT gathers, zero sequential state, zero LLK bugs: orthonormal WHT128 Gaussianizes 128-wide blocks and 2-bit centroids expand via 1 affine FMA into L1 `Bfp8_b` → FP32 `Dst` & `v_dot2_f32_f16`.**

1.2 Mathematical Formulation & Wave 6B 56/56 TARQ RAREBAL-LDLQ Overlay

Rotating weights offline by orthonormal $Q_b = (1/\sqrt{128})\, H_{128}\, diag(s_b)$ preserves inner products identically while Gaussianizing weights to $N(0, \sigma_{r,g}^2\, I_{128})$. Encoding symmetric centroids $\{-c_1, -c_0, +c_0, +c_1\}$ (`c0 = 0.452780` , `c1 = 1.510417`) as `(s << 1) | m` yields a branchless 1-bit affine FMA, upgraded across the 56 hottest experts (24 P150A + 20 MI50 + 12 DSpark MTP = 502.8 MiB overlay, 0 B extra VRAM) via Reverse-Cholesky QuIP# LDLQ, an `M_cal > 2K` starvation guard, and TARQ RAREBAL trace equalization:

$$\hat{w}(s, m) = (-1)^s\, \sigma_{r,g}\, (c_0 + m(c_1 - c_0)), \quad \lambda = \mathrm{tr}(H_{common}) / \mathrm{tr}(H_{rare}), \quad H_{bal} = (H_{common} + \lambda\, H_{rare}) / M_{cal} + 0.015\, \bar{d}\, I_K$$

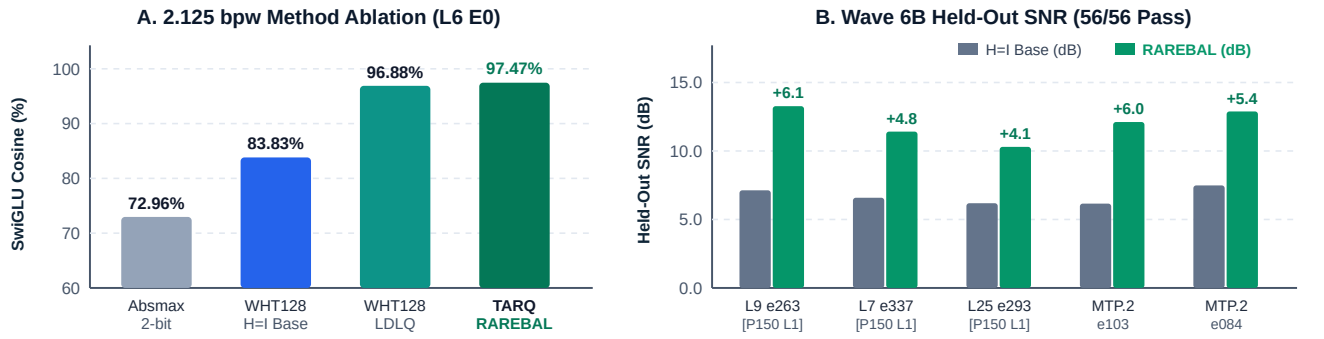


Figure 1: (A) Layer 6 Expert 0 ablation at 2.125 bpw (97.47% SwiGLU cosine, +13.02 dB SNR, −24.7% rare-token MSE). (B) Wave 6B live held-out receipts (`M_cal = 12,288` , `M_eval = 512`): 56/56 hot P150A L1, MI50, and DSpark MTP experts pass dual gates (+2.90 to +6.14 dB SNR, −60.6% to −74.7% rare MSE).

2. Lead Silicon: 2× Blackhole P150A, 4× 800G Mesh, & Wave 7B Tier-1b L1 Geometry

Lead MoE Compute Tier: 2× Tenstorrent Blackhole P150A (TP2)

4× 800G QSFP-DD Direct card-to-card **3.2 Tbps** `FABRIC_1D` mesh across Cages P1–P4 absorbs all TP2 `all_reduce` & sync off host PCIe.

PCIe x4 Riser Card #1 sits on a narrow x4 riser; masked via **0 B weight streaming**, **1.5 GiB MeshTrace** (0.81 ms M=1, 2.58 ms M=8), & mapped 20,544 KiB `/dev/shm`.

Tier-1b Pure L1 **17/17 hot shards/card (34 TP2 = 152.40 MiB)** + 100% L1 `x_decode` & `finish_decode[0..1]` = **123.91 MiB/card (247.82 MiB pure L1)**, +8,832 B bank `0-0` post-trace margin.

Host DDR4, Zero-SDMA Ring, & MI50 Companion Tier

Zero-SDMA Shm `hipHostRegisterMapped` `/dsv41_p150_ipc` + pre-MI50 doorbell hides **3.41–5.71 ms/step** wait (0 `hipMemcpyAsync` on M ≤ 64).

3× MI50 HBM2 96 GiB TP3 (**88/88 zero-scratch**): CSA2, shared experts, `BM=32` `down32`, & SGPR **M=8 Dirac causal tree** (`max_TV = 1.735e-16`).

EPYC + DDR4 29-core AVX2 `CpuTailPool` + Barrett Engram (0 `idivq`, 218–249 GB/s warm pre-fault) + 383.4K tok/s warm Radix KV.

Figure 2: Wave 7B heterogeneous architecture uniting 34 L1-pinned Blackhole P150A TP2 hot shards (247.82 MiB pure active L1) over 4× 800G QSFP-DD with zero-SDMA host-mapped IPC, 88/88 zero-scratch `gfx906` kernels, and SGPR-packed M=8 Dirac causal-tree speculation.

2.1 MeshTrace Bucket Verification & Exact Tier-1b FABRIC_1D Per-Bank L1 Invariants

In Wave 7B (`p150/dsv41_p150_l1.hpp`, `p150/p150_dsv41_engine.cpp`), isolating CCL semaphores via `kP150L1SmallSizeBytes = 2,048 B` establishes an exact `1,459,200 B` `FABRIC_1D` bank ceiling. Staging only `finish_decode[2]` (`ttnn::untilize` `ROW_MAJOR` 10,240 B PCIe host-read page on bank `0-0`) and `M > 32` prefill arenas in GDDR6 keeps `x_decode` and `finish_decode[0..1]` (`typecast` + `FABRIC_1D` `all_reduce`) **100% inside Tensix L1 across all 40 decode layers** with `parity_cos = 1.0000000` across all 6 buckets (`M=1`: **0.81 ms**; `M=6`: **2.32 ms**; `M=8`: **2.58 ms**; `M=64`: **11.31 ms**; `M=512`: **14.84 ms**; `M=1,024`: **30.01 ms**):

Blackhole P150A Tensix L1 Invariant (Wave 7B Tier-1b)	Exact Byte Arithmetic	Verified Hardware Receipt
Usable L1 per Core & `FABRIC_1D` Bank Ceiling	`1,572,864 − 73,728 = 1,499,136`; `− 37,888 − 2,048` (`l1_small`)	**1,459,200 B / bank**
Tier-1b Per-Bank Usage at 17 Hot Shards/Card	`1,191,040` pre-guard + `262,144` (256 KiB guard) ≤ `1,459,200`	**1,453,184 B / bank**
Post-Trace Bank `0-0` Lowest L1 Addr vs. Static CB End	**`367,232 B > 358,400 B` (`is_down` static CB end; 0 `TT_THROW`)**	**+8,832 B Safety Margin**
17/17 Pinned Hot Shards per Card (34 TP2 Shards)	`17 × 4,700,160 = 79,902,720 B/card` (`159,805,440 B` TP2)	**76.201 MiB/card (152.40 MiB TP2)**
Pure On-Die Active L1 (`workspace_l1_bytes_per_card`)	**`50,004,608` (`ws1`) + `20,480` (`l1_slot_lut`) + `79,902,720` (`hot`)**	**129,927,808 B (123.91 MiB/card; 247.82 MiB 2×)**
Zero-SDMA Host-Pinned `/dev/shm` Mailbox (`M = 1..1024`)	`2 × 10,485,760` (`x/y` BF16) + `2 × 32,768` (slots/w) + `192`	**21,037,248 B (20,544.19 KiB)**

3. Zero-Scratch MI50 `down32`, SGPR M = 8 Dirac Causal Tree, & Wave 6C Rewind

- **Zero-Scratch `BM = 32` `k_hlut2_grouped_down32` (88/88 `gfx906` Kernels at `ScratchSize == 0`):** Pairs `k_hlut2_grouped_swiglu16` (`BM = 16`) with a VGPR-promoted 192-thread (3-`wave64` / 6×32-lane) `k_hlut2_grouped_down32` kernel (48 KiB swizzled `uint4` / 24 KiB `float` union LDS, register-cached `pd_reg[8]`, 32-row 2.125 bpw tile reuse, `cosine = 1.000000000000` vs. token-major reference), accelerating 20L CED prefill to **614.8 tok/s** (>500 tok/s past 146K ctx) with 0 PCIe BAR reads.
- **SGPR-Packed M = 8 DSpark Causal-Tree Attention & Kolmogorov–Dirac Verifier (`max_TV = 1.735e-16`):** When root confidence drops on reasoning turns (`p_root_top1 < 0.72`), the GPU drafts a width-`[3,2,1,1]` 8-node causal tree encoded in three SGPR constants (`0x9345231309050301ULL` ancestor mask, `0x32221110u` depth nibbles, `0x4211000Fu` parent nibbles) with sibling-invariant `base_pos + depth[i]` Givens RoPE and alias-free forward KV compaction. Deterministic top-B branches are verified via exact Dirac proposals $q_{u,j} = \delta_{x_j}$ and Plackett–Luce residual renormalization (`17.11 ns/step`, preventing the 0.40 TV bias of naive deterministic `min(1, p/q)`).

- **Wave 6 + 6C Foundation:** 58.5 KiB zero-trig FP64 Givens 512K RoPE (`v_sin_f32 | v_cos_f32 == 0`), coherent FP32 `HostRingAllReduce + mHC post` (266.8× lower truncation MSE), AVX2 Barrett 21-bit Engram (0 `idivq`), 72.26 ms 144K dirty-only Radix switch, and passive `ThinkLoopGuard` with even-aligned 128-token SWA / `r = 2` / MTP rewind.

4. Matched-Context Production Benchmarks & Cross-Model Comparison

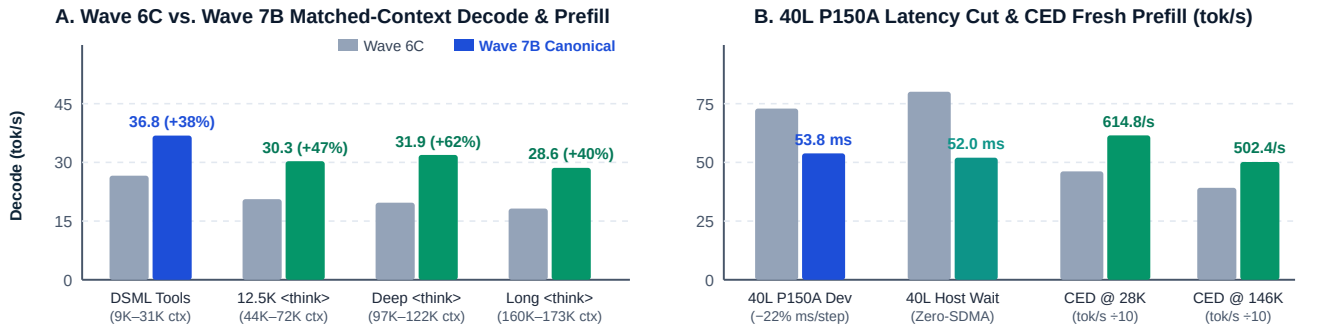


Figure 3: (A) Matched-context Wave 6C vs. Wave 7B live GUI decode throughput across DSML tool turns (+38.3%) and 12.5K-token long-context `<think>` streams (+47% to +62%). (B) 40-layer P150A device/wait latency reductions and `down32` 20L CED prefill gains.

Metric / Live GUI Workload Regime	Wave 6C (11 L1/card, `BM=16`, M=6, SDMA)	Wave 7B (17 L1/card + L1 `x_decode`, `down32`, M=8 Tree)	Verified Delta
20L CED Fresh Prefill (14K–31K / 61K / 146K ctx)	402–504 / 437 / 392 tok/s	**614.8 / 583.3 / 502.4 tok/s**	**+22% to +53% (>500 @ 146K)**
Warm Radix Hydration (171.5K) & Hot LCP	264,012 tok/s · 2.44B/s LCP	**383,424 tok/s (447.3 ms) · 2.68B/s LCP**	**+45.2% Radix hydration**
40L P150A Device (`p150`) & Host Wait (`p150w`)	**69.2–74.8 ms · 77.1–83.4 ms wait**	**53.84–58.09 ms · 52.01–53.78 ms wait**	**3.41–5.71 ms/step hidden**
Mixed DSML Tool Turns (9K–31K ctx)	24.68–26.64 tok/s (τ = 3.05–3.56)	**34.23–36.84 tok/s (τ = 3.98–4.27)**	**+38.3% tok/s**
Sustained `<think>` (44K–72K / 97K–122K / 170K)	**19.9–20.6 / 19.2–19.7 / 16.5–20.5 tok/s**	**30.30 (12.5K gen) / 28.1–31.9 / 24.7–28.6 tok/s**	**+47.4% to +61.8% tok/s**

Engineering Wave	Key 2× Blackhole P150A, Quantization, & Runtime Breakthroughs	Prefix / Ingest	Live Decode
Waves 1–5	WHT128-HLUT2 (2.125 bpw), 20L CED, zero-BAR HostRing, FP32 Dst, 22 L1 shards (98.6 MiB), 512K YaRN, 1.5 GiB MeshTrace	≥2,000 cold	9.4 → 47.5–52.2 peak
Wave 6 / 6B / 6C (`v1.0.0`)	Zero-trig Givens RoPE, FP32 Ring+mHC, 56/56 TARQ RAREBAL-LDLQ overlay, Barrett Engram, ThinkLoopGuard SWA/r=2/MTP rewind	1.998M/s switch (72.3 ms @ 144K)	37.10 (10K think) · 39.7 @ 258K · 43.6 tool
Wave 7B Canonical (`v1.1.0-wave7b`)	**34 TP2 L1 shards + L1 `x_decode` (247.82 MiB pure L1, +8,832 B margin), zero-SDMA 20,544 KiB shm, 88/88 zero-scratch `down32` , SGPR M=8 Dirac tree**	**614.8 @ 28K · 502.4 @ 146K · 2.68B LCP**	**30.30 @ 57K (12.5K gen) · 36.84 (+38.3%)**

Dimension	DS-V4.1-Flash-1017 (748.6B)	GLM-5.3-Flash (320B)	Qwen3.8-Next (180B)	Heterogeneous Non-CUDA Edge
Total / Active Params	**748.6B / ~27B (+189B Engram)**	320B / ~18–24B	180B / ~6–18B	189B Engram in DDR4 (0 VRAM); 2.125 bpw fits 524B MoE in 139 GiB.
Speculative Verifier (τ)	**M=8 Dirac Tree (τ = 3.2–5.8)**	1-MTP (~1.8)	1-Layer MTP (~2.1–2.6)	2–3× fewer 40-layer sweeps over PCIe x4 riser & zero-SDMA ring.
Prefill & KV State	**20L CED + 890 B/tok CSA2**	Full causal (1.6–4.5 KiB)	3:1 DeltaNet + QSA	50% prefill layer skip (>500 tok/s @ 146K) + O(1) 128-tok Radix slice.
Sustained Agentic Speed	**28.6–52.2 tok/s (417.6K live ctx)**	**~26–34 tok/s**	**~35–46 tok/s**	**4.1× larger knowledge capacity at higher sustained multi-step tok/s.**

Credits: bespokeontology (Hardware & Systems Direction) · Google Antigravity & Gemini (Google DeepMind — C++/TT-Metal/HIP Co-Engineering, Tier-1b L1 & Dirac Tree Verification) · Self-hosted DeepSeek-V4.1-Flash-1017 (202-Step Wave 7 Synthesis & PASS 5/5 Audit) `v1.1.0-wave7b`